

IAL Physics Unit 2 — Resistance, Temperature & Superconductivity | Recall Pack

IAL Physics Unit 2 — Resistance, Temperature & Superconductivity (incl. Core Practical 7)

The whole topic in one mental model. Resistance is just $R = V/I$ — voltage divided by current, *never* multiplied. A component obeys **Ohm's law** only while its **temperature is held constant** (straight I - V line through the origin, constant gradient $1/R$). The moment temperature changes, R changes — and the direction depends on the material: in a **metal** more heat means more lattice vibration, more electron-ion collisions, lower drift velocity, so R *ris*es (positive temperature coefficient, PTC). In a **semiconductor / NTC thermistor** more heat frees more charge carriers, so R *fall*s (negative temperature coefficient) — the exact opposite. Push the temperature the other way, far down, and a **superconductor** snaps to $\rho = 0$ exactly below its **critical temperature** T_c : zero resistance, zero heating, current that flows forever. Three materials, three different R - T stories on one graph.

1. Jargon glossary

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Resistance R	How hard a component fights the current — divide the voltage across it by the current through it. Bigger $R \Rightarrow$ smaller current for the same push.	The ratio of the potential difference across a component to the current through it: $R = V/I$. A property of the <i>specific component</i> — it depends on the material, its dimensions, and its temperature.	R ; ohms $\Omega = \text{V A}^{-1}$	NOT $V \times I$; NOT a fixed material constant (depends on geometry + temperature); NOT a store of electrons.	Lesson 11:31 — Marta: “resistance is voltage divided by the current.” One ohm = 1 V/A.	Phuc trap (lesson 44:01): Phuc wrote “ $R = I \times V$ ”. WRONG. $R = V/I$ — you <i>divide</i> , never multiply. Write the $V / I / R$ triangle: V on top.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Ohm's law	Current is proportional to voltage — <i>as long as the component doesn't heat up</i> . Double the voltage, double the current.	The current through a component is directly proportional to the p.d. across it, provided the temperature is constant : $I \propto V$. Gives a straight-line I - V graph through the origin and constant R .	$I \propto V$ (constant T)	NOT a universal law (only ohmic components obey it); NOT valid if the component heats up; the proportionality is $I \propto V$, NOT $I \propto V^2$.	Lesson 11:25 — straight line through origin = "ohmic conductor"; SaveMyExams 2-34.	Common trap: for full marks you MUST state <i>both</i> "directly proportional" AND "at constant temperature." Dropping the temperature condition loses the mark.
Ohmic conductor	A component whose resistance stays the same however much current you push through it — its I - V graph is a straight line.	A component for which V/I is constant (at constant T): obeys Ohm's law. Most metal wires and fixed resistors at low current.	straight I - V line; gradient = $1/R$	NOT a filament lamp or diode (those are non-ohmic); NOT constant if the wire is run hot.	Lesson 14:07 — Phuc confirms "ohmic = constant R = linear line."	Common trap: a metal wire is ohmic <i>only</i> if the current is small enough that it does not heat up. Large current $\Rightarrow$ heating $\Rightarrow$ non-ohmic.
Non-ohmic conductor	A component whose resistance changes as conditions change — its I - V graph is a curve, not a straight line.	A component for which V/I is not constant; the I - V characteristic is non-linear. Resistance must be read as $R = V/I$ at the point of interest, never from the gradient.	curved I - V ; $R = V/I$ at a point	NOT a straight line; resistance is NOT $1/\text{gradient}$ for a curve.	Lesson 12:17 — filament lamp curve; "if resistance is not constant, it will not obey Ohm's law."	Phuc trap (lesson 45:53): Phuc treated the current as <i>fixed</i> on a non-ohmic curve and tried $V = R \times I$ with the old I . WRONG — on a curve, I and V change together; use the graphical-intersection method.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Resistivity ρ	A property of the <i>material itself</i> , not the wire's shape — how strongly that substance resists current. Copper always has the same ρ whatever you make from it.	An intrinsic property of a material: $\rho = RA/L$. Depends on material and temperature, NOT on the dimensions of the sample. Low ρ = good conductor; high ρ = poor conductor.	ρ ; ohm-metres $\Omega \text{ m}$	NOT resistance (that depends on shape); NOT Ω/m — the unit is $\Omega \text{ m}$; NOT changed by cutting the wire.	Lesson 25:49 — “resistivity = how strongly a material resists current, regardless of use.”	Common trap: unit is $\Omega \text{ m}$ (ohm-METRES), not Ω/m . And ρ is a <i>material</i> property — bending or shortening the wire does not change ρ (it changes R).
Temperature coefficient (PTC / NTC)	A label for which way resistance moves when you heat the material: PTC = up (metals), NTC = down (semiconductors).	PTC (positive temperature coefficient): R increases with T — metals. NTC (negative temperature coefficient): R decreases with T — semiconductors / thermistors.	PTC: $R \uparrow$ as $T \uparrow$; NTC: $R \downarrow$ as $T \uparrow$	NOT the same sign for both materials; PTC is NOT semiconductors; NTC is NOT metals.	Textbook 10.4 — “copper has a positive temperature coefficient (PTC).”	Phuc trap (evidenced): metal vs semiconductor go <i>opposite</i> ways. Metal R rises with T ; NTC thermistor R falls with T . Do not mix them up — this is the #1 logged misconception.
NTC thermistor	A semiconductor sensor whose resistance <i>drops</i> as it gets hotter — the opposite of a metal. Used to sense temperature.	A negative-temperature-coefficient thermistor (a semiconductor). As T rises, the number density n of free charge carriers rises sharply, so $R = V/I$ falls steeply and non-linearly.	$R \downarrow$ as $T \uparrow$ (non-linear)	NOT a metal (metal does the reverse); NOT linear; NOT ohmic.	SaveMyExams 2-50 — thermistors in ovens, fire alarms, digital thermometers, thermostats.	Common trap: the mark-scheme reason is “ more charge carriers (free electrons) released ”, NOT “fewer collisions.” Quote carrier number, not collision frequency, for the NTC.

Term	Student-friendly meaning	Exact physics meaning	Symbols/units	What it is NOT	Lesson example	Phuc/common trap
Lattice vibration (metal mechanism)	The metal's positive ions jiggle on the spot; heat makes them jiggle harder, so flowing electrons crash into them more and slow down.	As T rises, the amplitude of vibration of the lattice ions increases $\Rightarrow$ more frequent electron-ion collisions $\Rightarrow$ reduced drift velocity v in $I = nAvq \Rightarrow$ lower I for the same $V \Rightarrow$ higher R .	qualitative; links to $I = nAvq$	NOT a change in carrier number n (for a metal n is fixed); NOT the thermistor mechanism.	Textbook analogy — running through 500 students who jump around (hot) vs stand still (cold).	Common trap: for a <i>metal</i> , n is constant; only v falls. Do not say “more carriers” for a metal — that is the semiconductor reason.
Superconductor	A material that, cooled below a special low temperature, loses ALL resistance — current flows with zero energy lost as heat, forever.	A material whose resistivity drops <i>abruptly to exactly zero</i> when cooled below its critical (transition) temperature T_c . Below T_c : $\rho = 0$, no heating, current persists indefinitely.	$\rho \rightarrow 0$ for $T < T_c$	NOT “very low resistance” — it is <i>exactly zero</i> ; NOT a normal conductor below T_c .	OneNote tile — $\rho \sim 10^{-18} \Omega \text{ m}$ below T_c (effectively zero); LK-99 clip.	Common trap: you MUST say “ exactly zero / no resistance. ” “Very low resistance” is NOT sufficient for the mark. State both <i>what</i> ($\rho = 0$) and the <i>condition</i> (below T_c).
Critical (transition) temperature T_c	The exact temperature at which a material suddenly flips into the superconducting state. Above it: normal. Below it: zero resistance.	The temperature below which a superconductor has zero resistivity. Above T_c it behaves as a normal conductor ($\rho > 0$); at T_c the resistivity drops discontinuously to zero.	T_c ; kelvin K	NOT a gradual change — it is a sharp phase transition; NOT room temperature for conventional superconductors.	Hg $T_c = 4.2 \text{ K}$; Nb 9.3 K; YBCO 92 K (high- T_c).	Common trap: label T_c on the R - T sketch as the point where R <i>drops vertically</i> to zero — not a gentle curve. Below T_c the line sits flat on $R = 0$.

2. Formula / case table

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Resistance (definition)	$R = \frac{V}{I}$	V = p.d. across component; I = current through it; R = resistance	R in Ω ; V in V; I in A	Defining resistance; finding R from a measured V and I ; reading R at a point on a non-ohmic curve.	$V = IR$; $I = \frac{V}{R}$	Phuc's #1 slip: it is V/I , NOT $I \times V$. You divide, never multiply.	$V = 6.0$ V, $I = 0.24$ A $\Rightarrow$ $R = 6.0/0.24 = 25$ Ω (self-test Q2).
Ohm's law (proportionality)	$I \propto V$ (constant T)	I = current; V = — p.d.; condition: temperature held constant	—	Stating the law; justifying a straight-line I - V graph through the origin.	$\$ \Rightarrow V/I = \$ \text{ const} \Rightarrow R \text{ const}$	Both halves needed: " $I \propto V$ " AND "at constant T ." Missing the T condition loses the mark.	Straight line through origin $\Rightarrow$ ohmic; gradient $= I/V = 1/R$.
Reading R from an I-V graph	Ohmic: $R = \frac{1}{\text{gradient}}$; Non-ohmic: $R = \frac{V}{I}$ at the point	gradient $= \Delta I / \Delta V$; for a curve, read V and I at the chosen point	Ω	Extracting resistance from a characteristic graph.	$R = 1/(\Delta I / \Delta V)$ (straight line only)	Phuc trap (45:53): do NOT use the gradient on a <i>curve</i> . Read off V and I and divide.	Lamp curve: at $V = 0.70$ V, $I = 40$ mA $\Rightarrow R = 0.70/(40 \times 10^{-3}) = 17.5$ Ω (lesson Q5a).
Non-ohmic graphical method	Draw line of gradient $1/R$ through origin; intersection with the curve gives (V, I)	gradient of the R -line $= I/V = 1/R$; intersection = operating point	V, A	Finding the p.d. that makes a non-ohmic component have a <i>target</i> R .	pick a test V : $I = V/R$, plot, draw line	Phuc trap (52:21): you cannot just multiply $R \times I$ with the old current — I is unknown; the <i>intersection</i> fixes both V and I .	For $R = 50$ Ω : pick $V = 0.60$ V $\Rightarrow I = 0.60/50 = 0.012$ A; line through origin cuts curve at $V \approx 0.60$ V (lesson Q5b).
Resistivity	$R = \frac{\rho L}{A}$	ρ = resistivity; L = length; A = cross-sectional area	ρ in Ω m; L in m; A in m^2	Relating a wire's resistance to its material and shape; finding ρ .	$\rho = \frac{RA}{L}$; $L = \frac{RA}{\rho}$; $A = \frac{\rho L}{R}$	L in m , A in m ² ; unit of ρ is Ω m (not Ω/m). Longer $\Rightarrow$ higher R ; thicker $\Rightarrow$ lower R .	$\rho = 1.7 \times 10^{-8}$, $L = 2.0$ m, $d = 0.50$ mm $\Rightarrow A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7}$, $R = \rho L / A = 0.17$ Ω .

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Area from diameter	$A = \pi \left(\frac{d}{2}\right)^2 = \frac{\pi d^2}{4}$	d = wire diameter; A = cross-sectional area	d in m; A in m^2	Computing A from a micrometer diameter for the resistivity formula.	$d = 2\sqrt{A/\pi}$	A 2% error in d gives a 4% error in A (and in ρ) because $A \propto d^2$.	$d = 0.50 \text{ mm} \Rightarrow A = \pi(0.25 \times 10^{-3})^2 = 1.96 \times 10^{-7} \text{ m}^2$.
Power dissipated	$P = IV = \frac{V^2}{R}$ $I^2 R = \frac{V^2}{R}$	P = power; I = current; V = p.d.; R = resistance	P in W	Power turned to heat in a resistor; choosing the form matching your known quantities.	$I = P/V$; $R = V^2/P$; $V = \sqrt{PR}$	Use the form with the quantities you know. $P = I^2 R$ derived by $V = IR$; $P = V^2/R$ by $I = V/R$.	$R = 47 \Omega$, $I = 85 \text{ mA} \Rightarrow P = I^2 R = (85 \times 10^{-3})^2 (47) = 0.34 \text{ W}$ (Example 6).
Fuse / safe-current check	$I = \frac{P}{V}$	P = power rating; V = supply voltage; I = operating current	A	“Show it works safely with an N A fuse” — compare operating I to the fuse rating.	$P = IV$; $V = P/I$	Rearrange to $I = P/V$, NOT $I = V/P$ (tutor’s own slip at 57:38, immediately corrected).	Kettle 240 V, 3.0 kW $\Rightarrow I = 3000/240 = 12.5 \text{ A} < 13 \text{ A}$ fuse $\Rightarrow$ safe (lesson Q6a).
Metal: R vs T (PTC)	$R(\theta) = R_0(1 + \alpha\theta)$	R_0 = resistance at 0°C ; α = temperature coefficient; θ = temperature in $^\circ\text{C}$	R in Ω ; θ in $^\circ\text{C}$; α in K^{-1}	Linear rise of metal resistance with temperature (CP7-style R - θ graph).	gradient = $R_0\alpha$; intercept = R_0 ; $\alpha = \text{gradient}/R_0$	Graph does NOT pass through the origin — intercept is R_0 , so it is linear but not proportional.	Copper: gradient/intercept $\Rightarrow \alpha \approx 4 \times 10^{-3} \text{ K}^{-1}$ (textbook Activity 10.3).
Semiconductor: R vs T (NTC)	$R \downarrow$ steeply (non-linear) as $T \uparrow$	n (carrier number density) rises with T ; $I = nAvq$ rises $\Rightarrow R = V/I$ falls	qualitative	Explaining/sketching a thermistor’s falling, curved R - T characteristic.	no linear formula — describe the trend)	Reason = “ more free carriers ”, NOT fewer collisions. Falls non-linearly, not linearly.	Cool a thermistor $\Rightarrow$ fewer carriers $\Rightarrow$ thermistor R rises $\Rightarrow$ series-circuit current falls (self-test Q5).

Quantity / case	Formula	Symbol meanings	Units	When to use	Rearrangements	Trap	Mini worked example
Superconductor below T_c	$\rho = 0$ (exactly) for $T < T_c$; $P = I^2 R = 0$	T_c = critical temperature; below it $\rho = 0$, $R = 0$, no heating	T_c in K	Superconductivity (phase transition definition, applications, and zero-loss arguments.	— no algebra)	Must say “ exactly zero ” + give condition “below T_c .” “Very low” is wrong. Above T_c it is a normal conductor.	Below T_c : lossless cables ($P = I^2 R = 0$), MRI/maglev magnets carry huge currents with no overheating.

3. R versus T — three materials at a glance

Material	Direction as $T \uparrow$	R - T shape	Microscopic reason	Mark-scheme phrase	Trap
Metal (e.g. copper)	R increases (PTC)	rising, $\sim$ linear, intercept R_0	More lattice vibration $\Rightarrow$ more electron-ion collisions $\Rightarrow$ lower drift velocity v (n fixed)	“amplitude of vibration of lattice ions increases $\Rightarrow$ more collisions $\Rightarrow$ reduced drift velocity”	n is constant for a metal — only v changes.
NTC thermistor (semiconductor)	R decreases (NTC)	falling steeply, non-linear	More free charge carriers released $\Rightarrow$ carrier number density n rises $\Rightarrow$ more current	“more charge carriers (free electrons) released $\Rightarrow$ resistance decreases”	Reason is <i>more carriers</i> , NOT <i>fewer collisions</i> . Opposite sign to metal.
Superconductor	$R = 0$ for $T < T_c$	normal above T_c , drops vertically to 0 at T_c , flat on zero below	Phase transition; ρ drops discontinuously to exactly zero	“resistivity drops to exactly zero below the critical temperature T_c ”	Must state “exactly zero” AND “below T_c .” Label T_c on the sketch.

4. Big picture (last-minute recall)

- $R = V/I$ — **divide** V by I , never multiply. (Phuc’s logged slip: $R = IV$ is WRONG.)
- Ohm’s law holds **only at constant temperature**: $I \propto V$, straight line through origin, R constant. Both halves needed for the mark.
- Non-ohmic curve: read $R = V/I$ at the point; never use the gradient. For a target R , use the graphical-intersection method (not the old fixed current).
- Resistivity $\rho = RA/L$ is a **material** property in $\Omega \text{ m}$; resistance depends on shape, ρ does not. $A = \pi d^2/4$ — a 2% error in d gives 4% in ρ .
- **Metal**: $T \uparrow \Rightarrow R \uparrow$ (more lattice vibration, lower drift velocity, n fixed) — PTC.

- **NTC thermistor:** $T \uparrow \Rightarrow R \downarrow$ (more free carriers) — NTC. **Opposite to the metal** — the key trap.
- **Superconductor:** below T_c , $\rho = 0$ *exactly*, no heating, $P = I^2 R = 0$ — lossless cables, MRI magnets, maglev, particle accelerators.
- Power: $P = IV = I^2 R = V^2/R$. Fuse check: $I = P/V$ (e.g. kettle $3000/240 = 12.5 \text{ A} < 13 \text{ A}$).
- Prefix trap: $40 \text{ mA} = 40 \times 10^{-3} \text{ A}$, NOT $40 \times 10^{-6} \text{ A}$. m = milli = 10^{-3} ; μ = micro = 10^{-6} .